A-level

ENVIRONMENTAL SCIENCE

Questions by topic: Ecological monitoring & new technologies - Set 1

Mark scheme

AQA spec 3.1.2.3.5-3.1.2.3.6; The living environment; textbook pp. 402-407

Marking guidance

• Each part opens with a line in bold saying how its marks are shared out, followed by one sentence on how to mark it.
• The figure in bold at the start of a marking point is the mark that point earns. Lines indented beneath it, starting with a dash, are the detail that belongs to the same mark; a line starting **OR** is another route to it.
• **Also accept** lists other wordings and answers that earn the mark - wording never has to match these notes, only the science. **Do not credit** lists the answers that look right and earn nothing.
• Where a question asks for a fixed number of points (eg *two* reasons), credit only the first points given.
• For levels-of-response questions, determine the level from the descriptors first, then position the mark within the level using the indicative content.
• Calculations: credit correct working (method marks) even if the final answer is wrong; incorrect answers with no working score 0.

Textbook references: every question below cites the printed page numbers in R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018) from which the content is drawn, together with the AQA 7447 specification reference.

Do not write outside the box

Question	Marking guidance	Marks
01.1	**2 marks per point** 2 marks for each of the 4 points below; award 1 mark where a point is only partly made. **2 Size:** shows whether the population is viable / how urgent action is – sets a baseline to judge success – needed to estimate carrying capacity and sustainable off-take **2 Distribution:** shows which habitats / areas are used and whether the population is fragmented into non-viable isolated groups needing corridors – identifies key areas to protect **2** Survival rate (mortality): identifies the life stage and causes of death (predation, disease, poaching) so management targets the limiting factor **2 Age structure:** many young indicates a growing population – few young / ageing population indicates breeding failure or high juvenile mortality even if adults are numerous – k-selected species need breeding adults protected **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **carrying capacity** → the largest population the area can support • **sustainable** → meeting present needs without stopping future generations meeting theirs • **limiting factor** → the factor in shortest supply that holds the rate or population back **Do not credit** • 'sustainable means it lasts a long time' - it means use that can continue without reducing what is available in future • 'the population cannot grow any more' as a definition - numbers fluctuate around the carrying capacity	8
01.2	**1 mark each** 1 mark for each of the 2 points below. **1** Direct counts from vehicles / aircraft / drones along transects, or camera traps with individual recognition **1** Sub-sample counts in known areas scaled up to the whole reserve – mark-release-recapture (Lincoln index) for smaller species **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **mark-release-recapture** → estimating population size from the proportion of marked animals recaptured	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 393-394, 400-402; AQA spec 3.1.2.3.5.	

Do not write outside the box

Question	Marking guidance	Marks
02.1	**2 marks per point** 2 marks for each of the 4 points below; award 1 mark where a point is only partly made. **2 Tracking:** continuous data on movements, territory size, migration routes and habitat use without disturbance, over large areas and long periods – acoustic tags in seawater – geolocators for small birds – more representative than occasional sightings **2 DNA:** identifies individuals, gene pool size / inbreeding and regional populations (tracing ivory, timber, fish) – eDNA detects rare or hidden aquatic species (great crested newt, invasive carp) from shed cells without capture **2 Image recognition:** software identifies individuals from unique markings (tiger stripes, whale shark spots, fin damage) in camera-trap / drone / satellite images, giving population size, lifespan, social structure from large image sets quickly and objectively **2 Acoustic:** bat detectors, hydrophones and sonograms record species that are hard to see (bats, cetaceans, insects), giving presence, abundance and activity continuously, day and night **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **validity** → actually measures what it is meant to measure • **gene pool** → the range of alleles / genes present in a population	8
02.2	**1 mark each** 1 mark for each of the 2 points below. **1** Data collected continuously / over large areas / at night rather than at a few times and places a researcher can visit **1** Less disturbance so animals behave naturally – larger sample sizes – objective rather than subjective identification **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 402-406; AQA spec 3.1.2.3.6.	
03.1	**Correct answer only** 1 mark for the correct option. No mark if more than one box is ticked. **1** B - radio waves do not travel through seawater – acoustic or satellite tags transmit position	1

Do not write outside the box

Question	Marking guidance	Marks
03.2	**2 marks per point** 2 marks for each of the 2 points below; award 1 mark where a point is only partly made. **2** Shows survival, dispersal and breeding of released individuals so success can be assessed and future releases improved **2** Radio / GPS tags or collars – leg rings / wing tags on birds – camera traps – DNA from droppings to confirm identity and diet **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **captive breeding** → breeding in captivity to increase numbers for release **Do not credit** • captive breeding alone as conservation - credit needs reintroduction, habitat protection or genetic management as well	4
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 393-394, 402-407; AQA spec 3.1.2.3.5-3.1.2.3.6.	
04.1	**Correct answer only** 1 mark for the correct option. No mark if more than one box is ticked. **1 A:** Environmental DNA (eDNA) analysis	1
04.2	**1 mark each** 1 mark for each of the 2 points below. **1** Transmitters are too heavy for small birds / their batteries would not last for a long-term study **1** Geolocator tags are small and only record time and light levels – latitude and longitude for each day are worked out when the bird is recaptured and the data downloaded **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
04.3	**1 mark each** 1 mark for each of the 2 points below. **1** Each separate (regional) population has its own unique gene pool **1** DNA from seized ivory can be compared with the database to identify where the elephant came from, so poaching hotspots / trade routes can be targeted **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **gene pool** → the range of alleles / genes present in a population	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 404-405; AQA spec 3.1.2.3.6.	

Do not write outside the box

Question	Marking guidance	Marks
05.1	**1 mark each** 1 mark for each of the 3 points below. **1** Images / maps of large areas – land-use change, deforestation, flooding **1** Vegetation density and surface temperature **1** Wind velocity, sea-surface altitude, wave height, oil pollution **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • 'trees give us the oxygen we breathe' as the main impact - credit carbon storage, soil protection, the water cycle and habitat loss	3
05.2	1 mark for the point below. **1** Microwaves pass through clouds (unlike visible light) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	1
05.3	**1 mark each** 1 mark for each of the 2 points below. **1** Very large / remote areas can be surveyed quickly and repeatedly so changes over time are detected **1** No need for researchers to visit inaccessible areas, reducing cost and disturbance – data are standardised across the whole area **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • Points after the first two - the question asks for two, so a list of extras cannot pick up the marks. • 'trees give us the oxygen we breathe' as the main impact - credit carbon storage, soil protection, the water cycle and habitat loss	2
05.4	**Any 4 of 5** 1 mark each for any 4 of the 5 points below, in any order, maximum 4. **1** Motion-sensitive cameras placed along trails record animals passing, using visible light by day and infrared at night **1** Each tiger has a unique pattern of facial / body stripes **1** Images are compared with the database so individuals are recognised rather than counted twice **1** The number of different individuals recorded gives a minimum population – repeated captures give territory size, movements, lifespans and social groupings **1 Limitation:** only works for species with unique visible features **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **infrared** → heat radiation / long-wave radiation re-emitted by the Earth	4
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 402-403, 406-407; AQA spec 3.1.2.3.6.	

Do not write outside the box

Question	Marking guidance	Marks
06.1	**1 mark each** 1 mark for each of the 2 points below. **1** Its reading is adjusted to match a meter known to be accurate so that the results are accurate / comparable **1** If it cannot be adjusted the percentage error should be recorded so the results can be corrected **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **accurate** → close to the true value	2
06.2	**Any 3 of 6** 1 mark each for any 3 of the 6 points below, in any order, maximum 3. **1 Droppings:** diet, gender, territory size **1 Owl pellets:** diet **1** Feeding marks (eg chewed hazel nuts): presence of species such as dormice **1 Nests / burrows:** presence and breeding **1 Tracks / footprints:** presence, movements **1** Territorial marks (scratching posts): presence and territory **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • Points after the first three - the question asks for three, so a list of extras cannot pick up the marks.	3
06.3	**1 mark each** 1 mark for each of the 2 points below. **1** Bat detectors / sonograms record the ultrasonic sounds bats produce, which differ between species **1** The recordings provide evidence of presence, abundance and activity of species that cannot easily be seen **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
06.4	**Any 3 of 4** 1 mark each for any 3 of the 4 points below, in any order, maximum 3. **1** Marked individuals mix freely with the rest of the population **1** Marks do not fall off or wear away, and marking does not affect survival **1** The population does not change during the study (no births, deaths or migration) **1** All individuals have an equal chance of being caught / animals are mobile and can be caught **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **mark-release-recapture** → estimating population size from the proportion of marked animals recaptured **Do not credit** • Points after the first three - the question asks for three, so a list of extras cannot pick up the marks.	3

Do not write outside the box

Question	Marking guidance	Marks
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 402, 405, 407; AQA spec 3.1.2.3.5-3.1.2.3.6.	
07.1	**Calculation** A correct final answer scores full marks on its own. If the answer is wrong, each correct step of working below earns its mark. **1** (15 - 6) / 6 x 100 **1** = 150% **Also accept** • The correct answer however it is set out, including a different but valid method. • An answer carried forward from an earlier incorrect step (error carried forward). • An answer rounded to 2 or more significant figures, and the answer written as a fraction, decimal or percentage where the question does not specify. • The value alone - the unit (%) is already printed on the answer line. **Do not credit** • A bare wrong answer with no working.	2
07.2	**1 mark each** 1 mark for each of the 2 points below. **1** Cells shed by the newts into the water contain their DNA **1** Water samples are analysed for DNA sequences unique to the species **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
07.3	**1 mark each** 1 mark for each of the 2 points below. **1** Catches depend on trapping effort, weather and newt activity on the night, so they vary from year to year **1** Trapping only samples one pond and one part of the population, whereas eDNA surveys all 20 ponds with a standard method **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **reliable** → repeatable / gives consistent results when repeated **Do not credit** • 'repeat the readings to make them more accurate' - repeats improve reliability; accuracy depends on the method and the instrument	2

Do not write outside the box

Question	Marking guidance	Marks
07.4	**Levels of response** Decide the level from the descriptors first, then use the indicative content to place the mark within that level. The points below are not a checklist. **Level 3 (7-9 marks)**: Answer is comprehensive, relevant and logically structured. Shows detailed knowledge and understanding of the relevant scientific ideas and applies them accurately to the context. Evaluation/analysis is well supported by evidence and a clear judgement or conclusion is reached. **Level 2 (4-6 marks)**: Answer is mostly relevant and shows some structure. Knowledge and understanding of relevant ideas is sound but application to the context or evaluation may be incomplete or lack detail. A judgement may be made but is not fully supported. **Level 1 (1-3 marks)**: Answer is basic, with limited structure. Some relevant knowledge is shown but there are omissions or errors. Little application to the context and little or no evaluation. **0 marks**: No relevant content. **Indicative content - a guide to placing the mark, not a checklist** • Image databases of unique markings (tiger stripes, whale shark spots, dolphin fins) giving territory size, movements, lifespan and social groups – limited to species with unique features • Motion-sensitive and infrared cameras, CCTV of nests with less disturbance – marking (rings, wing tags) • **DNA:** eDNA to detect presence (great crested newts, invasive carp) – DNA databases to identify regional populations and trace timber, ivory and fish • **Auditory monitoring:** bat detectors and sonograms for presence, abundance and activity • **Position monitoring:** radio, GPS and acoustic transmitters – geolocator tags for small birds – data recorders – carrier systems - ROVs / drones, AUVs, balloons, aircraft, satellites, animals • **Satellites:** visible light for land use and deforestation, infrared for vegetation and temperature, radar for wind, waves, sea level and oil – GPS for tracking • **Evaluation:** greater coverage, less disturbance and continuous data vs cost, battery / weight limits, calibration, need for ground-truthing, ethical issues of tagging – indirect evidence and traditional methods still needed • Justified conclusion **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • Any accurate, relevant material that is not in the indicative content. • **infrared** → heat radiation / long-wave radiation re-emitted by the Earth	9
	Do not credit • 'trees give us the oxygen we breathe' as the main impact - credit carbon storage, soil protection, the water cycle and habitat loss	
	Model answer - Level 3 (9 marks)	

Do not write outside the box

Question	Marking guidance	Marks
	New technologies have transformed what can be monitored and how often. Image databases of unique markings - tiger stripes, whale-shark spots, dolphin fin scars - let individuals be recognised, giving territory size, movements, lifespan and social structure without repeated capture, though only for species with unique features. Motion-sensitive and infrared cameras and CCTV record activity day and night with far less disturbance than observers, especially at nests.	
	Genetics adds detection where nothing can be seen: eDNA in water reveals great crested newts or invasive carp from shed cells, and DNA databases identify which regional population timber, ivory or fish came from, supporting enforcement. Bat detectors and sonograms give presence, abundance and activity of species heard but not seen. Position monitoring uses radio, GPS and acoustic transmitters, or tiny geolocator tags for birds too small to carry a transmitter, while data recorders on drones, AUVs, balloons, aircraft, satellites or the animals themselves collect abiotic data. Satellites map land use and deforestation in visible light, vegetation density and temperature in infrared, and wind, waves, sea level and oil slicks with radar that works through cloud.	
	The gains are coverage, continuity and reduced disturbance. The limits are cost, battery life and weight, the need to calibrate sensors and ground-truth remote data, ethical questions about tagging, and the fact that many species still require traditional methods and indirect evidence. New technology therefore complements rather than replaces fieldwork, but for rare, remote or nocturnal species it has made effective conservation possible where it was not before.	
	Weaker answer - Level 1 (3 marks)	
	New technologies help monitor wildlife. Cameras can take pictures of animals at night. GPS collars can track where animals go. Satellites can see deforestation from space. DNA can tell which animals are present. Bat detectors can hear bats. These help conservation because we know where animals are and how many there are. They can be expensive though.	
	What makes the difference	
	• Level 1 lists technologies with no detail of how they work or what data they give; Level 3 links each to the information it provides (territory size, presence, activity).	
	• Level 3's evaluation considers several limitations (calibration, ground-truthing, ethics, species coverage) and reaches a balanced judgement; Level 1 has one word - 'expensive'.	
	• Structure: Level 3 groups by imagery, genetics, sound, position and remote sensing; the list in Level 1 has no organisation.	
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 402-407; AQA spec 3.1.2.3.5-3.1.2.3.6.	

Do not write outside the box